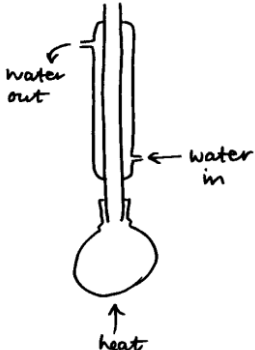


| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                  | Marks available |           |          |           |          |          |
|----------|-----|------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----------|----------|-----------|----------|----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                  | AO1             | AO2       | AO3      | Total     | Maths    | Prac     |
|          | (c) | (i)  |  | ethanol / ethanal and acidified potassium dichromate (1)<br>colour change from orange to green (1)<br>OR<br>ethanol / ethanal and acidified potassium manganate(VII) (1)<br>colour change from purple/light pink to colourless (1)                                                                                               |                 | 2         |          | 2         |          | 2        |
|          |     | (ii) |  | <br>condenser with water in at bottom and out at top (1)<br>condenser attached upright to top of reaction vessel (1)<br>source of heat (with upright condenser drawn) (1)<br>if distillation apparatus drawn – (1) max for water in / water out |                 |           | 3        | 3         |          | 3        |
|          |     |      |  | <b>Question 10 total</b>                                                                                                                                                                                                                                                                                                         | <b>0</b>        | <b>10</b> | <b>6</b> | <b>16</b> | <b>8</b> | <b>9</b> |